

DICOM Metadata Extractor

Program Documentation

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1 Overview

The **DICOM Metadata Extractor** is a Python utility designed to read a DICOM file and generate a formatted text report containing all header information (tags, names, and values). This output is intended to be easily copied or imported into the *Meta Data PET* tab of the Radcom PET Report Template, where users can view or document key acquisition parameters and scanner metadata for QA or recordkeeping purposes.

The program simplifies the process of collecting detailed metadata without requiring specialized DICOM viewer software. It can read nearly any valid DICOM file, even those that may contain unusual or non-standard tag lengths.

2 Purpose

The primary goal of this tool is to allow medical physicists to quickly extract and document PET/CT study metadata in a standardized text format. This is especially useful for:

- Identifying acquisition and reconstruction parameters for PET quality control.
- Verifying scanner calibration factors and SUV computation metadata.
- Archiving scanner information for ACR or regulatory documentation.
- Facilitating troubleshooting by comparing DICOM header data across studies or systems.

3 Program Workflow

3.1 Step 1: File Selection

Upon execution, a graphical file selection dialog appears, allowing the user to choose a single DICOM file (`.dcm`) to analyze. The dialog filters for valid DICOM extensions but also allows selection of any file type for flexibility in mixed exports.

If no file is selected, the program displays a message indicating that the process was canceled.

3.2 Step 2: Output Directory Selection

After the DICOM file is chosen, a second dialog prompts the user to select an output directory. This is where the text file containing the extracted metadata will be saved. If the user cancels this step, the program again exits gracefully.

3.3 Step 3: DICOM Metadata Extraction

The script reads the selected DICOM file using the `pydicom` library with `force=True`, allowing it to handle files that may not strictly adhere to the DICOM standard (e.g., private tags,

corrupted headers).

Each DICOM element is processed as follows:

1. The tag number is extracted and formatted as a hexadecimal pair, e.g., (0010,0010).
2. The element name (e.g., *PatientName*, *SeriesDescription*) is recorded.
3. The value is converted to a human-readable string. Multi-value fields are concatenated with commas.
4. Line breaks and tabs are removed for a clean table-style output.
5. Values longer than 100 characters are truncated with an ellipsis (...) for readability.

The final output is stored as a three-column table with the following header:

Tag	Name	Value
-----	------	-------

Each subsequent line corresponds to one DICOM attribute, producing a comprehensive listing of all available metadata.

3.4 Step 4: Text File Generation

The formatted metadata text is automatically saved to the selected output directory using the following naming convention:

```
[OriginalFilename]_metadata_[YYYYMMDD]_[HHMMSS].txt
```

This ensures that multiple exports can coexist without overwriting prior results.

After saving:

- A pop-up message confirms success and displays the full file path.
- On Windows systems, the file automatically opens in the default text editor for immediate review.

If any error occurs (e.g., file access denied, invalid directory), a descriptive error dialog is displayed.

4 Implementation Details

4.1 Core Functions

- **select_dicom_file()**: Prompts user to select a single DICOM file.
- **select_output_directory()**: Prompts user to choose a directory to save the output.
- **extract_dicom_metadata(file_path)**: Reads all DICOM elements and returns a formatted string of tag-name-value triples.
- **save_to_txt(text, original_file_path, output_dir)**: Writes metadata to a timestamped text file and opens it for review.

4.2 Key Libraries

- **pydicom**: Handles reading and parsing of DICOM datasets.
- **tkinter**: Provides GUI dialogs for user interaction.
- **datetime**: Used to timestamp exported text files.
- **re (Regular Expressions)**: Cleans tag formatting and removes invalid characters.

5 Output Formatting for Report Template

The generated text is intentionally designed to paste directly into the *Meta Data PET* tab within the Radcom Report Template. Each line follows a consistent delimiter pattern (‘—’) which allows for:

- Clean alignment when pasted into Excel.
- Easy parsing or lookup using Excel functions (e.g., TEXTSPLIT or FILTER).
- Consistent readability across platforms.

An example output snippet:

```
Tag | Name | Value
(0008,0020) | Study Date | 20251016
(0008,0030) | Study Time | 142500.000000
(0008,0060) | Modality | PT
(0010,0010) | Patient Name | DOE^JOHN
(0028,0030) | Pixel Spacing | 2.73438, 2.73438
```

6 How to Use the Program

1. Run `2._extract_dicom_to_clipboard.py`.
2. When prompted, select a DICOM file for analysis.
3. Choose an output folder to save the text file.
4. Review the automatically opened text file.
5. Copy the full text content and paste it into the *Meta Data PET* tab.

7 Error Handling and Limitations

- The program assumes that the selected file is a valid DICOM dataset; non-DICOM files may return incomplete or unreadable data.
- Multi-value and binary tags (e.g., pixel data) are excluded or truncated for brevity.
- On macOS/Linux, the automatic open feature (`os.startfile`) may not execute; users can manually open the saved text file.

8 Conclusion

The **DICOM Metadata Extractor** streamlines documentation and analysis of PET/CT metadata by providing a structured, human-readable export compatible with the Radcom PET Report Template. By consolidating the full header content into a standardized format, the tool reduces manual transcription effort and enhances QA traceability.

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